9-1

$$q=\frac{t_{f1}-t_{f2}}{\frac{1}{h_1}+\frac{\delta_1}{\lambda_1}+\frac{\delta_2}{\lambda_2}+\frac{\delta_3}{\lambda_3}+\frac{1}{h_2}}=\frac{30-(-2)}{\frac{1}{2}+\frac{0.0008}{45}+\frac{0.15}{0.07}+\frac{0.07}{0.1}+\frac{1}{3}}=\boxed{10.2\ W/m^2}$$

$$Q=qSt=10.2(3600)A=36720\cdot A\ J$$

9-4

$$q=\frac{t_{w1}-t_{w2}}{\frac{\delta_1}{\lambda_1}}=\frac{t_{w2}-t_{w3}}{\frac{\delta_2}{\lambda_2}}$$

$$\frac{45-23}{\frac{0.015}{0.15}}=\frac{23-18}{\frac{0.016}{\lambda_2}}\qquad \lambda_2=0.704\ W/m\cdot K$$

9-8

$t_{w1}=600°C$ $\quad t_{w2}=500°C$ $\quad t_{w3}=250°C$ $\quad t_{w4}=50°C$

$$q=\frac{t_{w1}-t_{w2}}{R_1}=\frac{t_{w2}-t_{w3}}{R_2}=\frac{t_{w3}-t_{w4}}{R_3}$$

$$\frac{100}{R_1}=\frac{250}{R_2}=\frac{200}{R_3}\qquad R_1=\frac{\delta_1}{A_1\lambda_1}\quad R_2=\frac{\delta_2}{A_2\lambda_2}\quad R_3=\frac{\delta_3}{A_3\lambda_3}$$

$$\frac{R_1}{R_2}=\frac{100}{250}\qquad \frac{R_2}{R_3}=\frac{5}{4}$$

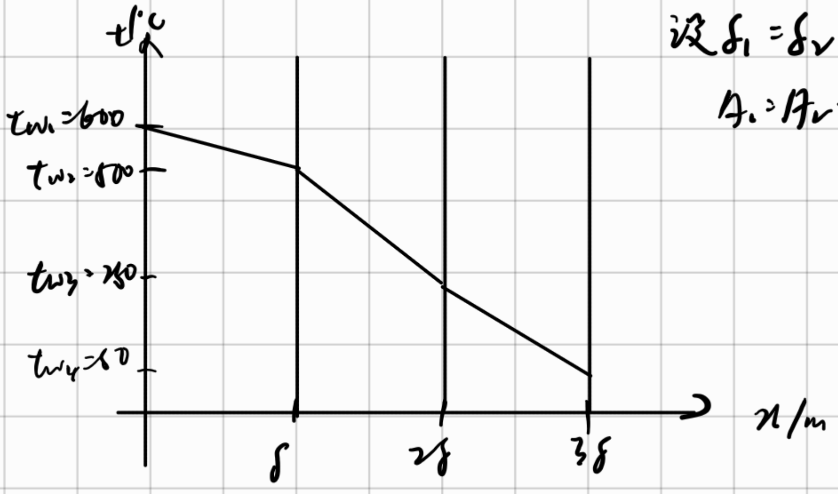
$$\lambda_1>\lambda_3>\lambda_2$$

$$\left|\frac{dt}{dx}\right|_1>\left|\frac{dt}{dx}\right|_3>\left|\frac{dt}{dx}\right|_2$$

$$\frac{R_1}{R_2}=\frac{2}{5}\qquad R_1:R_2:R_3=2:5:4$$

设 $\delta_1=\delta_2=\delta_3=\delta$

$A_1=A_2=A_3=A$



9-7

(1) $R_1 = \frac{1}{2\pi\lambda_1} \ln \frac{d_2}{d_1} = \frac{1}{2\pi(45)} \ln \frac{160}{150} = 2.2826 \times 10^{-4}\ K \cdot m/W$

$R_2 = \frac{1}{2\pi\lambda_2} \ln \frac{d_3}{d_2} = \frac{1}{2\pi(0.1)} \ln \left(\frac{160+2(40)}{160}\right) = 0.6453\ K \cdot m/W$

$R_3 = \frac{1}{2\pi\lambda_3} \ln \frac{d_4}{d_3} = \frac{1}{2\pi(0.16)} \ln \left(\frac{240+2(50)}{240}\right) = 0.3465\ K \cdot m/W$

(2) $\Phi = \frac{t_{w1}-t_{w4}}{R_1+R_2+R_3} = \frac{400-50}{0.992} = 352.8226\ W/m$

(3) $\Phi = \frac{t_{w1}-t_{w2}}{R_1} = \frac{t_{w2}-t_{w3}}{R_2} = \frac{t_{w3}-t_{w4}}{R_3}$

$t_{w2} = t_{w1} - \Phi R_1$
$= 399.9195°C$

$t_{w3} = \Phi R_3 + t_{w4}$
$= 172.253°C$